

Edge Geography

is XNLP-hard for Pathwidth

and in XP for Tree-Partition Width

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01 The game

Two players move one token along unused edges. Each move deletes the edge just used. The first player unable to move loses.

Can Player 1 force a win?

Both variants are PSPACE-complete in general. We ask whether a graph’s width can make winner determination tractable.

02 What graph structure buys us

PATHWIDTH

XNLP-hard

Directed and undirected
Edge Geography.

Also: Directed Edge Geography is FPT for treewidth + maximum degree, with time $f(k, \Delta) n^{O(1)}$.

03 Why pathwidth is hard

Source problem. In *Chosen Maximum Outdegree*, orient weighted edges so that each vertex respects its outgoing-weight bound.

Choose. Player 1 encodes an orientation by choosing a side of each edge gadget.

Challenge. Player 2 tests whether a vertex exceeds its bound. A feasible orientation exists exactly when Player 1 can force a win.

FIGURE SPACE

Orientation choices and the challenge gadget

The construction bounds the new pathwidth by a function of the source width. A constant-size gadget then transfers the result to undirected graphs.

Pathwidth: overlapping vertex bags arranged in a path; width = largest bag size minus one.

OPEN QUESTION

Is either variant in XP when parameterized by pathwidth?

XNLP-hardness rules out an FPT algorithm unless $FPT = XNLP$; it does not rule out XP.

FIGURE SPACE

A graph and a short sequence of moves

TREE-PARTITION WIDTH

XP

Both variants on simple graphs.
Running time $n^{f(k)}$.

04 Why tree partitions help

Partition the vertices into bags arranged in a tree. Edges stay inside a bag or join adjacent bags. Each bag has at most k vertices.

On simple graphs, a parent–child cut has at most k^2 edges: the **ports** through which play crosses the boundary.

FIGURE SPACE

Bags, ports and equivalent child subtrees

Summarize. An interface type records entry and return behavior, the player to move, and the remaining type after an excursion.

Count. Children of the same type are interchangeable. Store their multiplicities and process the bags bottom-up.

For fixed k , the type universe is finite. The compressed local games yield a total running time of $n^{f(k)}$.